

Algebra 1
Unit 2
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

1. True or False? *The polynomial $5xy^7 + 3x^4y^2 - x^2y^2$ is written in standard form.*

True. The highest degree is written first then the terms are in order from highest degree to lowest degree.

2. Write the polynomial $-10xy^5 + 2x^3y^2 - 6x^2y^2$ in standard form.

$$-10xy^5 + 2x^3y^2 - 6x^2y^2$$

For questions 3-4, rewrite each polynomial in standard form, then identify the features of the polynomial.

3. $4y^2 - 8y^3 + 2 - 6y$

Standard Form: $-8y^3 + 4y^2 - 6y + 2$

Number of Terms: 4

Degree of the polynomial: 3

Leading Term: $-8y^3$

Leading Coefficient: -8

4. $-3mp^2 + 7m^2p^2 - 4p - 6m^4p^2$

Standard Form: $-6m^4p^2 + 7m^2p^2 - 3mp^2 - 4p$

Number of Terms: 4

Degree of the polynomial: 6

Leading Term: $-6m^4p^2$

Leading Coefficient: -6

5. Create a polynomial that has 3 terms, a degree of 6, a leading coefficient of 5, and a constant term of -1.8 .

$$5x^6 + 4x^2 - 1.8 \quad \text{Answers will vary.}$$

6. Alexis found the product of $3x^2(5x^4 - 3x^3 + 7x - 8)$ to be $15x^8 - 9x^6 + 21x^3 - 8x^2$. Is she correct? Explain.

No, she multiplied rather than adding exponents and multiplied the last term incorrectly.

7. Find the product of $-2r^2(12r^3 - 2r^2 - 9r + 11)$ in standard form.

$$-24r^5 + 4r^4 + 18r^3 - 22r^2$$

Find the product of each by distributing the monomial to the polynomial. Express the product in standard form.

8. $\frac{1}{2}y^2(6y^2 - 6y + 7)$

$$3y^4 - 3y^3 + 3.5y^2$$

9. $(6xy^2 + 4x^2y + 7xy + 8)(\frac{2}{3}x^3y^2)$

$$4x^4y^4 + \frac{8}{3}x^5y^3 + \frac{14}{3}x^4y^3 + \frac{16}{3}x^3y^2$$

10. $-2.8a^3(1.2a^4 - 2.3a^3 + 0.6a)$

$$-3.36a^7 + 6.44a^6 - 1.68a^4$$